

Excess-Degree Bounds for Minimal Counterexamples to the Erdős–Gyárfás Conjecture

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Version 1 - August 2026

Abstract

Suppose that the Erdős–Gyárfás conjecture has a counterexample, and choose one first with minimum order and then minimum size. Let A be its degree-three vertices, let B be its vertices of degree at least four, put $b = |B|$, and set $s = \sum_{v \in B} (d(v) - 4)$. We prove

$$|A| \geq 2b + s + 4.$$

Consequently $3|A| \geq 2|V| + s + 4$ and $|E| \leq 2|V| - b - 2$. The proof records every two-edge B – A_1 – B branch in an auxiliary graph on B . The absence of a 4-cycle makes this graph simple, exact cycle lifting and minimality make it 2-degenerate, and an edge-cut identity plus parity supplies the final unit. The cases $b = 0$ and $b = 1$ require separate square-free arguments. This conditional theorem strengthens the closest known degree-distribution bound but does not prove or disprove the conjecture. Standard-library exact checks provide finite falsification tests; the written argument is the proof.

Keywords. Erdős–Gyárfás conjecture; minimal counterexample; degree sequence; degeneracy; prescribed cycle lengths.

Mathematics Subject Classification (2020). 05C38; 05C35; 05C75.

1 Introduction

The Erdős–Gyárfás conjecture asks whether every finite graph of minimum degree at least three contains a cycle whose length is a power of two [2]. It remains open. Computation and restricted-class theorems have supplied substantial evidence and structure, but finite frontiers do not settle the global question; see, for example, Markström [4] and the recent cubic-bipartite result of Tranquilli [6].

Carr [1] recently studied a counterexample chosen first with minimum order and then minimum size. He proved that every vertex is adjacent to a vertex of degree three, that the vertices of degree at least four form an independent set, and that, with $A = V_3$ and $B = V_{\geq 4}$,

$$3|A| \geq 4|B|.$$

Thus at least $4/7$ of the vertices are cubic. The purpose of this note is to give an excess-sensitive strengthening:

$$|A| \geq 2|B| + \sum_{v \in B} (d(v) - 4) + 4.$$

The proof is short but uses more than the direct bipartite edge count. It encodes certain two-edge branches through cubic vertices in an auxiliary graph, transfers the forbidden cycle lengths to that

graph, and invokes minimality a second time to obtain 2-degeneracy. An exact cut identity and parity then yield the additive constant.

The surrounding critical-graph framework is classical. Erdős, Faudree, Gyárfás, and Schelp [3] initiated the study of graphs without proper subgraphs of minimum degree three. Narins, Pokrovskiy, and Szabó [5] clarified the induced versus non-induced settings and proved, in the non-induced equality case, a pancyclic structure theorem. In particular, their results already imply the prior general bound $|E| \leq 2|V| - 3$ for a minimal Erdős–Gyárfás counterexample. The contribution claimed here is only the exact excess-sensitive inequality and its auxiliary-graph proof. No claim of absolute historical priority is made.

2 Minimality and notation

All graphs are finite, simple, and undirected; every cycle is simple. A *dyadic cycle* has length 2^k for some integer $k \geq 2$. Assume that a counterexample exists, and let G minimize the pair $(|V(G)|, |E(G)|)$ lexicographically among all graphs of minimum degree at least three with no dyadic cycle. Write

$$A = \{v : d_G(v) = 3\}, \quad B = \{v : d_G(v) \geq 4\}, \quad b = |B|,$$

and

$$s = \sum_{v \in B} (d_G(v) - 4).$$

For $0 \leq i \leq 3$, put

$$A_i = \{x \in A : d_{G[A]}(x) = i\}, \quad a_i = |A_i|.$$

Lemma 1. *Every proper subgraph of G has minimum degree at most two. Every vertex of G has a neighbor in A , and B is independent. In particular, $a_0 = 0$.*

Proof. If a proper subgraph H had minimum degree at least three, then H would also have no dyadic cycle. Fewer vertices would contradict minimum order, while the same vertex set and fewer edges would contradict secondary minimum size.

For $v \in V(G)$, the proper subgraph $G - v$ has a vertex x of degree at most two. Since deleting v lowers any degree by at most one and $\delta(G) \geq 3$, we have $xv \in E(G)$ and $d_G(x) = 3$. Hence every vertex has an A -neighbor. If an edge joined two vertices of B , deleting it would leave minimum degree at least three, contradicting the first paragraph. The neighbor property applied to vertices of A gives $a_0 = 0$. $\square$

3 The auxiliary branch graph

Every $x \in A_1$ has exactly two neighbors in B . Define an auxiliary graph F with vertex set B by recording one edge between these two neighbors for each $x \in A_1$. This is a defined branch graph; it is not asserted to be a subgraph, minor, quotient, or literal suppression of G .

Lemma 2. *The graph F is finite and simple, $|E(F)| = a_1$, and every simple r -cycle in F lifts to a simple $2r$ -cycle in G .*

Proof. Simplicity of G makes the two B -neighbors of an A_1 vertex distinct, so no loop is created. If distinct vertices $x, y \in A_1$ defined the same pair $u, v \in B$, then u, x, v, y, u would be a 4-cycle. Thus branch representatives are injective, F is simple, and $|E(F)| = a_1$.

Now take a simple cycle in F . Its edges are distinct, hence their branch representatives in A_1 are distinct. The cycle vertices lie in B , so they are also distinct from all representatives. Replacing each auxiliary edge by its defining two-edge branch therefore gives a simple cycle of exactly twice the length in G . $\square$

Lemma 3. *The graph F is 2-degenerate. If $b \geq 2$, then*

$$a_1 \leq 2b - 3.$$

Proof. By Lemma 2, a dyadic cycle in F would lift to a dyadic cycle in G . Thus F is dyadic-cycle-free. If a nonempty subgraph $J \subseteq F$ had minimum degree at least three, then J itself would be a counterexample. Lemma 1 makes A nonempty, so $|V(J)| \leq b < |V(G)|$, contradicting minimum order. Hence F is 2-degenerate.

In a degeneracy ordering of a simple graph on $b \geq 2$ vertices, each of the first $b - 2$ vertices has at most two forward edges, the penultimate vertex has at most one, and the last has none. Therefore $|E(F)| \leq 2(b - 2) + 1 = 2b - 3$, and Lemma 2 completes the proof. $\square$

4 The excess bound

Because B is independent, every edge incident with B crosses the cut (A, B) . A vertex of A_i has exactly $3 - i$ neighbors in B , and $a_0 = 0$. Counting the cut at both endpoints gives the exact identity

$$4b + s = e(A, B) = 2a_1 + a_2. \quad (1)$$

Theorem 4 (Excess-degree bound). *For the lexicographically minimal counterexample G ,*

$$\boxed{|A| \geq 2b + s + 4.}$$

More precisely, if $b \geq 2$, then $a_2 \geq s + 6$; if $b = 1$, then $a_2 = 4 + s$, $a_3 \geq 2\lceil(4 + s)/2\rceil$, and $|A| \geq 8 + 2s + (s \bmod 2)$; and if $b = 0$, then $|A| \geq 8$.

Proof. First suppose $b \geq 2$ and put

$$\delta = 2b - 3 - a_1 \geq 0$$

using Lemma 3. Equation (1) gives

$$a_2 = s + 6 + 2\delta \quad (2)$$

and hence

$$|A| = a_1 + a_2 + a_3 = 2b + s + 3 + \delta + a_3. \quad (3)$$

The degree sum of $G[A]$ is $a_1 + 2a_2 + 3a_3$, so $a_1 + a_3$ is even. Since $a_1 = 2b - 3 - \delta$ and $2b - 3$ is odd, $\delta + a_3$ is odd. It is a nonnegative integer and therefore at least one. Equations (2) and (3) prove the assertions for $b \geq 2$.

Now let $b = 1$, and let v be the unique vertex of B . An A_1 vertex would require two distinct B -neighbors, so $a_1 = 0$ and Equation (1) gives $a_2 = 4 + s$. Put $S = A_2 = N_G(v)$ and $T = A_3$. Every vertex of S has degree two in $G[A]$ and at most one neighbor in S , since two such neighbors together with v would form a 4-cycle. Thus every vertex of S has a neighbor in T and $e(S, T) \geq |S|$. Conversely, a vertex of T has at most one neighbor in S , by the same square obstruction. Hence

$$|T| \geq e(S, T) \geq |S| = 4 + s,$$

and $|A| \geq 8 + 2s$. The degree sum of $G[A]$ is $2a_2 + 3a_3$, so a_3 is even. Therefore

$$a_3 \geq 2 \left\lceil \frac{4+s}{2} \right\rceil \quad \text{and} \quad |A| \geq 8 + 2s + (s \bmod 2).$$

Finally, let $b = 0$. Then G is simple cubic, so its order is even. The unique cubic graph on four vertices is K_4 , which has a 4-cycle. On six vertices, the complement of a cubic graph is 2-regular and hence is either C_6 or two disjoint triangles. The corresponding cubic graphs are the triangular prism and $K_{3,3}$, both of which have a 4-cycle. Therefore a dyadic-cycle-free cubic graph has order at least eight. $\square$

Corollary 5. *Under the hypotheses of Theorem 4, with $n = |V(G)|$ and $m = |E(G)|$,*

$$\boxed{3|A| \geq 2n + s + 4} \quad \text{and} \quad \boxed{m \leq 2n - b - 2}.$$

If $b = 1$, the sharper exceptional estimate is

$$\boxed{m \leq 2n - 4 - \lceil s/2 \rceil}.$$

Proof. Since $n = |A| + b$, the first inequality is a rearrangement of Theorem 4. The degree sum is

$$2m = 3|A| + 4b + s,$$

and therefore

$$4n - 2m = |A| - s \geq 2b + 4.$$

Division by two gives the edge bound.

For $b = 1$, the exceptional estimate in Theorem 4 gives $|A| - s \geq 8 + s + (s \bmod 2) = 8 + 2\lceil s/2 \rceil$. Since $4n - 2m = |A| - s$, division by two proves the sharper formula. $\square$

Remark 6 (Equality regimes). *For $b \geq 2$, equality in Theorem 4 holds exactly when, with $\delta = 2b - 3 - a_1$, either $(\delta, a_3) = (1, 0)$ or $(0, 1)$. This follows immediately from Equation (3) and the fact that $\delta + a_3$ is positive and odd. The stronger exceptional estimates show that $b = 0$ and $b = 1$ cannot attain the central equality.*

Remark 7 (Prior density boundary). *The general conclusion $m \leq 2n - 3$ is already implicit in the classical critical-graph literature [5]. Corollary 5 recovers that value when $b = 1$ and is strictly stronger when $b \geq 2$. We do not claim the statement “average degree below four” as new.*

5 Sharpness, verification, and scope

The counting mechanism cannot increase the constant four using only the local ingredients above. The supplementary appendix records a 16-vertex graph with $b = 4$, $s = 0$, and $(a_0, a_1, a_2, a_3) = (0, 4, 8, 0)$. Its auxiliary graph is a triangle with a pendant edge, and it attains $|A| = 2b + s + 4$. The graph contains dyadic cycles, so it is not a counterexample and does not challenge the theorem. It shows only that a stronger constant requires additional global information or a different invariant.

A standard-library exact checker exhausts 343,980 bounded admissible integer tuples, all 33,866 labelled simple auxiliary graphs on two through six vertices, 281,197 exact cycle lifts, and all labelled cubic graphs on four and six vertices. A method-distinct replay uses integer edge masks and precomputed cycle masks and reproduces every core aggregate. Six intentional mutations

are rejected. These checks are finite corroboration, not a proof for arbitrary order; the preceding argument is the proof.

The theorem is conditional on the existence of a counterexample. It does not prove that a counterexample exists, that none exists, that B is empty, or that a minimal counterexample is cubic or 3-connected. No two-port, multipole, or finite terminal-path classification is used. A current, claim-specific literature search located no exact or equivalent statement of Theorem 4; this supports novelty only within the recorded search scope and does not establish absolute priority or external validation.

Data and code availability

The proof is self-contained. The public paper repository contains the exact checkers, tests, mutation controls, expected outputs, environment record, and SHA-256 manifests used for the release replay. The frozen mixed-rights private archive and excluded two-port, TP4–20, and CE22–26 materials are not redistributed. No public reuse license is granted.

AI-use disclosure

OpenAI model-assisted systems participated in proof reconstruction, adversarial checking, software development, source organization, literature search, and manuscript preparation. Automated systems are not authors or external reviewers. The originating proof and program were not treated as self-certifying. The named human author must inspect and approve every claim, citation, disclosure, and artifact before submission and retains responsibility for the work.

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